Project Design Phase-I
Solution Architecture

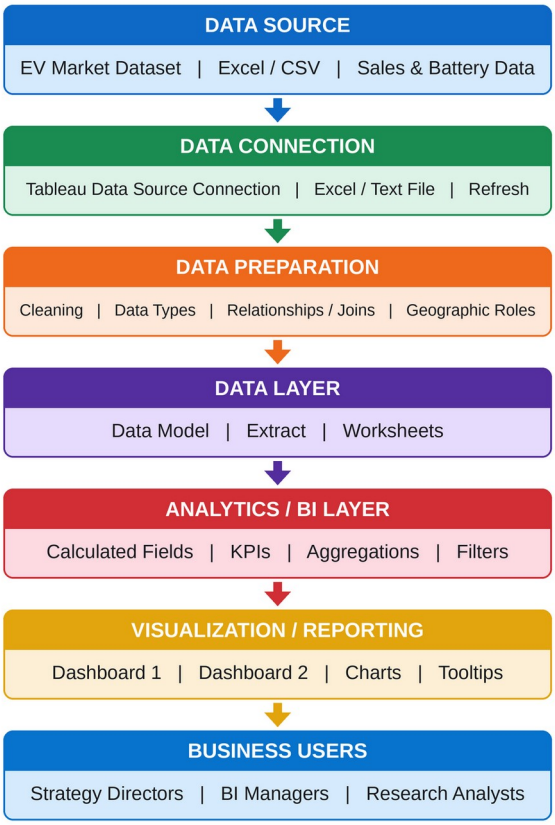
Date	19 September 2026
Team ID	PNT2022TMIDxxxxxx
Project Name	EV Market Performance and Insights
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Data Source	The EV market dataset that holds vehicle sales, price, battery, range, charging, manufacturer, country, rating and CO2 information.	Excel / CSV file
Data Connection	Connects Tableau to the dataset and lets the data be refreshed when the source file changes.	Tableau Desktop data source connection
Data Preparation	Cleans the data, sets data types, relates tables and assigns geographic roles for country and region fields.	Tableau Data Source tab
Data Layer	Stores the prepared data model and extract that every worksheet and dashboard reads from.	Tableau data model / extract
Analytics / BI Layer	Calculates KPIs and measures such as total sales, average price, average range, CO2 saved, retained value and efficiency, and applies filters and aggregations.	Tableau calculated fields and LOD expressions

Visualization / Reporting	Presents the insights as charts in two interactive dashboards with a shared category legend, filters and tooltips.	Tableau Desktop (Dashboard 1 and Dashboard 2)
Business Users	Strategy directors, BI managers and researchers explore the dashboards to compare segments, manufacturers and technologies and to make decisions.	Tableau dashboards (Tableau Public / browser)